STAT 405 **Project Proposal** **Manya Jain**
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Bayesian Analysis of Global Happiness Determinants

1. Team

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GitHub Repository: https://github.com/MANYAdsci/STAT405-project

2. Introduction

Understanding the factors that influence happiness across countries is an important question in economics and public policy. Governments increasingly use measures of subjective well-being to evaluate social progress beyond traditional economic indicators. In this project, I aim to investigate how economic, social, and institutional factors contribute to national happiness levels.

While the World Happiness Report uses standard regression methods to estimate the contribution of different factors to happiness, this project focuses on Bayesian inference to quantify the uncertainty in these relationships. In particular, hierarchical models will be used to account for regional variation in happiness levels.

3. Datasets

3.1 Main Dataset

The primary dataset is derived from the World Happiness Report dataset (2020–2024), which contains country-level measures of happiness and several economic and social indicators commonly used to study well-being across countries. For this project, only the 2023 and 2024 observations are used. These years are combined to increase the sample size and improve the stability of the regression estimates.

The variables used in the analysis include:

- happiness – reported happiness score (life evaluation)
- gdp – GDP per capita
- social_support – perceived social support
- life_expectancy – healthy life expectancy
- freedom – freedom to make life choices
- generosity – generosity index
- corruption – perception of corruption

The response variable is the happiness score, while the remaining variables serve as predictors.

A region variable was created using the countrycode package in R to group countries into world regions (e.g., Europe & Central Asia, East Asia & Pacific, Sub-Saharan Africa). This allows the analysis to account for regional variation using hierarchical models. The resulting dataset contains approximately 300 observations across seven global regions.

Data source:
https://plos.figshare.com/articles/dataset/Dataset_from_the_World_Happiness_Report_2020_2024_covering_social_and_economic_indicators_of_156_countries_and_regions_/28905777?file=54103710

Example structure of the dataset:

```
head(happiness_data)
```

A tibble: 6 × 11

overall_rank	country	happiness	gdp	social_support	life_expectancy	freedom	generosity	corruption	year	region
<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	Finland	7.632	1.305	1.592	0.874	0.681	0.202	0.393	2023	Europe & Central Asia
2	Norway	7.594	1.456	1.582	0.861	0.686	0.286	0.340	2023	Europe & Central Asia
3	Denmark	7.555	1.351	1.590	0.868	0.683	0.284	0.408	2023	Europe & Central Asia
4	Iceland	7.495	1.343	1.644	0.914	0.677	0.353	0.138	2023	Europe & Central Asia
5	Switzerland	7.487	1.420	1.549	0.927	0.660	0.256	0.357	2023	Europe & Central Asia
6	Netherlands	7.441	1.361	1.488	0.878	0.638	0.333	0.295	2023	Europe & Central Asia

```
tail(happiness_data)
```

A tibble: 6 × 11

overall_rank	country	happiness	gdp	social_support	life_expectancy	freedom	generosity	corruption	year	region
<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
151	Yemen	3.380	0.287	1.163	0.463	0.143	0.108	0.077	2024	Middle East & North Africa
152	Rwanda	3.334	0.359	0.711	0.614	0.555	0.217	0.411	2024	Sub-Saharan Africa
153	Tanzania	3.231	0.476	0.885	0.499	0.417	0.276	0.147	2024	Sub-Saharan Africa
154	Afghanistan	3.203	0.350	0.517	0.361	0.000	0.158	0.025	2024	South Asia
155	Central African Republic	3.083	0.026	0.000	0.105	0.225	0.235	0.035	2024	Sub-Saharan Africa
156	South Sudan	2.853	0.306	0.575	0.295	0.010	0.202	0.091	2024	Sub-Saharan Africa

3.2 Backup Dataset

As a backup dataset, we will use the **World Happiness Report 2023 dataset**, which contains the same variables for a single year.

This dataset includes the same key variables used in the analysis, such as happiness score, GDP per capita, social support, life expectancy, freedom, generosity, and corruption. Using a single-year dataset avoids potential inconsistencies in variable definitions across multiple years while still allowing the analysis of relationships between happiness and its determinants.

It comes from the same data source but only the data from 2023 is used.

```
head(backup_data)
```

A data.frame: 6 × 10

	overall_rank	country	happiness	gdp	social_support	life_expectancy	freedom	generosity	corruption	year
	<int>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>
1	1	Finland	7.632	1.305	1.592	0.874	0.681	0.202	0.393	2023
2	2	Norway	7.594	1.456	1.582	0.861	0.686	0.286	0.340	2023
3	3	Denmark	7.555	1.351	1.590	0.868	0.683	0.284	0.408	2023
4	4	Iceland	7.495	1.343	1.644	0.914	0.677	0.353	0.138	2023
5	5	Switzerland	7.487	1.420	1.549	0.927	0.660	0.256	0.357	2023
6	6	Netherlands	7.441	1.361	1.488	0.878	0.638	0.333	0.295	2023

4. Research Question

This project investigates the relationship between national happiness levels and several economic and social indicators. The primary research question is: **Which economic, social, and institutional factors are most strongly associated with national happiness, and how uncertain are these relationships?**

To address this, the analysis will estimate the posterior distributions of the regression coefficients for GDP per capita, social support, life expectancy, freedom, generosity, and perceptions of corruption. A secondary question is whether countries within the same world region share similar baseline happiness levels, which will be examined using a hierarchical Bayesian model with region-level effects.

5. Exploratory Data Analysis

Exploratory analysis was conducted to understand relationships between predictors and happiness.

Figure 1	Figure 2	Figure 3
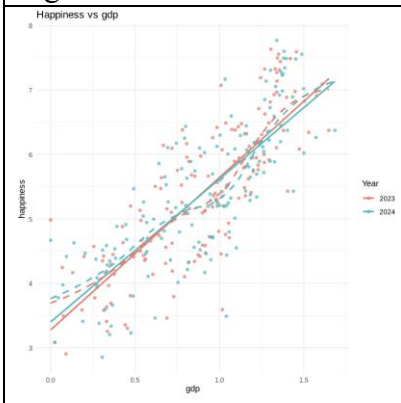	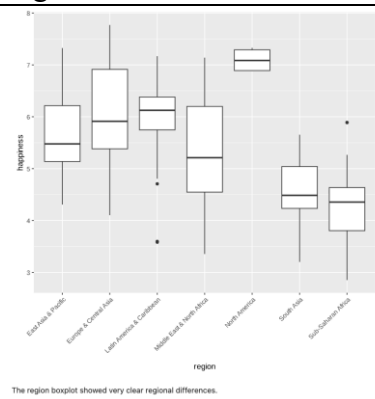	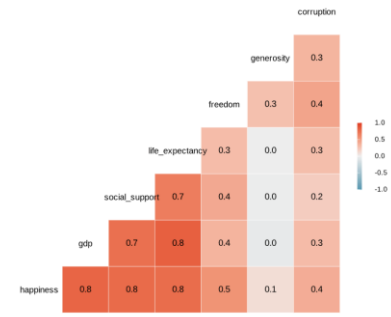
Figure 1 shows a strong positive relationship between GDP and happiness, suggesting that economic development is an important determinant of well-being.	Figure 2 illustrates the distribution of happiness across global regions. Clear regional differences are observed, with North America and Europe showing higher median happiness levels than South Asia and Sub-Saharan Africa.	Figure 3 presents the correlation matrix among the variables. GDP, social support, and life expectancy show the strongest correlations with happiness (approximately 0.75–0.80). Moderate correlations between predictors are present, but none exceed thresholds typically associated with severe multicollinearity.

These results suggest that a linear modeling framework is appropriate, while also motivating the inclusion of regional effects to account for shared regional characteristics.

6. Methodology

The analysis will use Bayesian regression models to estimate the relationship between national happiness levels and several economic and social indicators. Bayesian methods allow us to estimate the posterior distributions of model parameters, which provides a principled way to quantify uncertainty in the estimated effects of each predictor. Two models will be developed in order to compare different modeling assumptions.

6.1 Baseline Model

The first model is a Bayesian linear regression model, where the happiness score is modeled as a function of the economic and social indicators included in the dataset.

$$happiness_i = \beta_0 + \beta_1 gdp_i + \beta_2 social_support_i + \beta_3 life_expectancy_i + \beta_4 freedom_i + \beta_5 generosity_i + \beta_6 corruption_i + \epsilon_i$$

This model assumes that all countries share the same baseline level of happiness and that the relationships between predictors and happiness are identical across countries.

6.2 Extended Model

The second model introduces **regional variation** using a hierarchical structure. Countries are grouped into global regions, and each region is allowed to have its own baseline level of happiness.

$$happiness_{ij} = \beta_0 + \beta_1 gdp_{ij} + \beta_2 social_support_{ij} + \beta_3 life_expectancy_{ij} + \beta_4 freedom_{ij} + \beta_5 generosity_{ij} + \beta_6 corruption_{ij} + u_j + \epsilon_{ij}$$

Where $u_j \sim N\left(0, \sigma^2_{region}\right)$ represents a region-specific random effect capturing regional differences in baseline happiness levels.

This hierarchical model allows the analysis to account for similarities among countries within the same region.

6.3 Posterior Computation

Posterior distributions will be estimated using two Bayesian inference methods. First, Hamiltonian Monte Carlo (HMC) implemented in Stan via the **brms** interface in R will be used to obtain posterior samples. Second, an approximate method such as **variational inference or Laplace approximation** will be used to obtain a faster approximation to the posterior. The two approaches will be compared in terms of posterior estimates, uncertainty intervals, and computational efficiency.

6.4 Model Evaluation

The analysis will focus on the posterior distributions of the regression coefficients to identify which predictors are most strongly associated with national happiness. Model comparison will examine posterior means, credible intervals, and uncertainty in the estimated effects, as well as differences between the baseline and hierarchical models and between inference methods.

6.5 Planned Tools

- The analysis will be conducted in R using the following packages:
- tidyverse for data manipulation
- ggplot2 for visualization
- brms / rstan for Bayesian modeling
- posterior / bayesplot for posterior diagnostics and visualization

7. AI use Declaration

ChatGPT was used to assist with refining the wording, improving clarity of written explanations, and organizing sections of the proposal. All the research ideas, methods, dataset selection, and methodological decisions were developed and reviewed individually.

References

Helliwell, J. F., Layard, R., Sachs, J. D., De Neve, J.-E., Aknin, L. B., & Wang, S. (2023). *World Happiness Report data*. Sustainable Development Solutions Network. https://worldhappiness.report/data-sharing/

Helliwell, J. F., Layard, R., Sachs, J. D., De Neve, J.-E., Aknin, L. B., & Wang, S. (2023). *World Happiness Report 2023*. Sustainable Development Solutions Network. https://worldhappiness.report/

Yang, B., & Xie, X. (2025). *Analyzing and predicting global happiness index via Integrated Multilayer Clustering and machine learning models*. PLOS ONE. https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0322287#sec011